

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Barco
Firmware Version: N/A
Model(s): UDX-4K32, UDX-4K22

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.04.0001-b002	2.3.1

Version History

Module Version	Date	Notes
1_2_0_0	8/29/2019	Added Focus and Illumination commands. Removed unsupported Stereo 6P value from Desired Display Mode. Added Service and Error values to Power status. Removed port changeability. Fixed serial pin diagram.
1_1_0_0	10/19/2018	- Added UDX-4K22 model. - Renamed Input value 'L1 DisplayPort 1' to 'L1 DisplayPort' - Removed unsupported Input value 'L1 DisplayPort 2' - Added Input value 'L1 Quad SDI'
1_0_0_0	10/10/2018	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='UDX-4K32')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='UDX-4K32')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 9090, Model='UDX-4K32')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format with Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command DesiredDisplayMode	Value 'Mono' 'Night Vision'	Value 'Auto Stereo' 'IG Pixel Shift'	Value 'Active Stereo'
# DesiredDisplayMode example InterfaceName.Set('DesiredDisplayMode', 'Mono')			
Command Focus	Value 'In'	Value 'Out'	
# Focus example InterfaceName.Set('Focus', 'In')			
Command Illumination	Value 0 to 100 in steps of 1		
# Illumination example InterfaceName.Set('Illumination', 100)			
Command Input ¹	Value 'L1 DisplayPort' 'L1 HDBaseT 1' 'L1 SDI B' 'L2 DisplayPort A' 'L2 DisplayPort D' 'L2 Dual DP - BD' 'L2 Dual Head DP - B D' 'L2 Quad column DP'	Value 'L1 Quad SDI' 'L1 HDBaseT 2' 'L1 SDI C' 'L2 DisplayPort B' 'L2 Dual DP - AB' 'L2 Dual DP - CD' 'L2 Dual Head Dual DP - AB CD'	Value 'L1 HDMI' 'L1 SDI A' 'L1 SDI D' 'L2 DisplayPort C' 'L2 Dual DP - AC' 'L2 Dual Head DP - A C' 'L2 Quad DP'
# Input example InterfaceName.Set('Input', 'L1 DisplayPort')			
Command InvertImageStereoGlassSync	Value 'On'	Value 'Off'	
# InvertImageStereoGlassSync example InterfaceName.Set('InvertImageStereoGlassSync', 'On')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			

¹ All L2 Input values are present only when the optional Quad DP1.2 input slot is installed

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	Value
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value		
Illumination	0 to 100 in steps of 1		
# Illumination example InterfaceName.Update('Illumination') Value = InterfaceName.ReadStatus('Illumination') InterfaceName.SubscribeStatus('Illumination', None, FeedbackHandler)			
Command	Value	Value	Value
Input ¹	'L1 DisplayPort'	'L1 Quad SDI'	'L1 HDMI'
	'L1 HDBaseT 1'	'L1 HDBaseT 2'	'L1 SDI A'
	'L1 SDI B'	'L1 SDI C'	'L1 SDI D'
	'L2 DisplayPort A'	'L2 DisplayPort B'	'L2 DisplayPort C'
	'L2 DisplayPort D'	'L2 Dual DP - AB'	'L2 Dual DP - AC'
	'L2 Dual DP - BD'	'L2 Dual DP - CD'	'L2 Dual Head DP - A C'
	'L2 Dual Head DP - B D'	'L2 Dual Head Dual DP - AB CD'	'L2 Quad DP'
	'L2 Quad column DP'		
# Input example InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command	Value	Value	
InvertImageStereoGlassSync	'On'	'Off'	
# InvertImageStereoGlassSync example InterfaceName.Update('InvertImageStereoGlassSync') Value = InterfaceName.ReadStatus('InvertImageStereoGlassSync') InterfaceName.SubscribeStatus('InvertImageStereoGlassSync', None, FeedbackHandler)			
Command	Value	Value	Value
Power	'On'	'Off'	'Warming Up'
	'Cooling Down'	'Booting Up'	'Eco / Power Save Mode'
	'Service'	'Error'	

```
# Power example  
InterfaceName.Update('Power')  
Value = InterfaceName.ReadStatus('Power')  
InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)
```

¹ All L2 Input values are present only when the optional Quad DP1.2 input slot is installed

Cable and Adapter Requirements

Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 19200

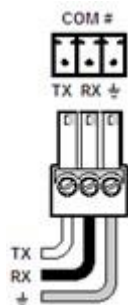
Data Bits: 8

Parity: None

Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type:	Ethernet
Default Port:	9090
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Desired Display Mode Active Stereo	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.display.desireddisplaymode", "value": "ActiveStereo"}}
Desired Display Mode Auto Stereo	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.display.desireddisplaymode", "value": "AutoStereo"}}
Desired Display Mode IG Pixel Shift	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.display.desireddisplaymode", "value": "IGPixelShift"}}
Desired Display Mode Mono	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.display.desireddisplaymode", "value": "Mono"}}
Desired Display Mode Night Vision	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.display.desireddisplaymode", "value": "NightVision"}}
Focus In	{"jsonrpc": "2.0", "method": "optics.focus.stepforward", "params": {"steps": 1}}
Focus Out	{"jsonrpc": "2.0", "method": "optics.focus.stepreverse", "params": {"steps": 1}}
Illumination 0	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "illumination.sources.laser.power", "value": 0.0}}
Illumination 100	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "illumination.sources.laser.power", "value": 100.0}}
Input L1 DisplayPort	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "L1 DisplayPort"}}
Input L1 HDBaseT 1	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "L1 HDBaseT 1"}}
Input L1 HDBaseT 2	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "L1 HDBaseT 2"}}
Input L1 HDMI	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "L1 HDMI"}}
Input L1 Quad SDI	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "L1 Quad SDI"}}
Input L1 SDI A	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "L1 SDI A"}}
Input L1 SDI B	{"jsonrpc": "2.0", "method": "property.set",

Global Scriptor Module Communication Sheet

	"params": {"property": "image.window.main.source", "value": "L1 SDI B"}}}
Input L1 SDI C	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "L1 SDI C"}}}
Input L1 SDI D	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "L1 SDI D"}}}
Input L2 DisplayPort A	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "L2 DisplayPort A"}}}
Input L2 DisplayPort B	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "L2 DisplayPort B"}}}
Input L2 DisplayPort C	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "L2 DisplayPort C"}}}
Input L2 DisplayPort D	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "L2 DisplayPort D"}}}
Input L2 Dual DP - AB	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "L2 Dual DP - AB"}}}
Input L2 Dual DP - AC	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "L2 Dual DP - AC"}}}
Input L2 Dual DP - BD	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "L2 Dual DP - BD"}}}
Input L2 Dual DP - CD	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "L2 Dual DP - CD"}}}
Input L2 Dual Head DP - A C	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "L2 Dual Head DP - A C"}}}
Input L2 Dual Head DP - B D	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "L2 Dual Head DP - B D"}}}
Input L2 Dual Head Dual DP - AB CD	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "L2 Dual Head Dual DP - AB CD"}}}
Input L2 Quad column DP	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "L2 Quad column DP"}}}
Input L2 Quad DP	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "L2 Quad DP"}}}

**Global Scripter Module
Communication Sheet**

	"image.window.main.source", "value": "L2 Quad DP"]}]}
Invert Image Stereo Glass Sync Off	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.stereo.glassync.invert", "value": false}}
Invert Image Stereo Glass Sync On	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.stereo.glassync.invert", "value": true}}
Power Off	{"jsonrpc": "2.0", "method": "system.poweroff", "params": {}}
Power On	{"jsonrpc": "2.0", "method": "system.poweron", "params": {}}

Appendix B. Update Commands

Illumination	<code>{"jsonrpc": "2.0", "method": "property.get", "params": {"property": "illumination.sources.laser.power"}}</code>
Input	<code>{"jsonrpc": "2.0", "method": "property.get", "params": {"property": "image.window.main.source"}}</code>
Invert Image Stereo Glass Sync	<code>{"jsonrpc": "2.0", "method": "property.get", "params": {"property": "image.stereo.glassync.invert"}}</code>
Power	<code>{"jsonrpc": "2.0", "method": "property.get", "params": {"property": "system.state"}}</code>

